

Mandatory Assignment:

Build bag-of-word model using term frequency from an existing document

I. INTRODUCTION

This is a mandatory assignment, a part of PGR210 Machine Learning and Natural Language Processing course. Students are required to preprocess four specified texts; a description of the city of Tromsø, Oslo, a description of a text mining course, and the entire E-book “Crime and Punishment”. Then build Bags of Words (BoW) in the same dimension for the four texts, after that, come up with a query string and perform a simple text analysis to identify its most similar text.

II. METHODOLOGY

To successfully complete the tasks, students must demonstrate the understanding of key principles in Natural Language Processing (NLP). The following are some of the methodologies required.

BoW Model: Getting the computers to understand human language is challenging for two main reasons. Firstly, machine learning primarily deals with numbers, whereas human language is non-numeric. Secondly, creating a model to comprehend all diverse human languages is a complex task.

The "bag of words" technique is a fundamental approach in machine learning for extracting and quantifying words from text. The word “bag” implies that there is no order or structure to the known words in the collection. A typical BoW Pipeline includes:

- **Step 1: Text acquisition:** Acquire the text to process. In this case, the texts were given.
- **Step 2: Preprocessing:** It is important to evaluate what steps are needed depending on, among other factors, the nature of the problems and the text data being processed. While the specifics steps to derive at the usable Bags-of-Words may vary, the common steps include the following.
 - **Tokenization:** The process of spitting the documents into a collection of smaller units which could be paragraph, sentences. In this case, we split the text into individual words or tokens.
 - **Lowercasing:** The method to change all capital letters to lower captions, making it easier for a computer to compare words and letters, as any capital letters has different representation than lowercase letters. All uppercase could have been used, but lower case is more practical as the NLTK stop words are all given in lower case.
 - **Removing white space:** In this step, all white spaces in each token are removed.
 - **Stop words Removal:** Stop words are words that do not meaningfully contribute to the understanding of the text such as “the”, “and” are removed. This can significantly reduce the size of the documents, particularly in the case of “Crime and Punishment” book. General rule of thumb is to use stop words in the same language of the text. In this case, English.
 - **Remove Punctuation and numbers:** While it is generally accepted that punctuations do not contribute to the overall meaning of the text, therefore, are removed. Dealing with numbers is a different matter. Whether or not to have them removed depends primarily on the nature of the texts and the purpose of the analysis and the model employed. Models that do not requires numbers such as *sentiment analysis* where sentiment is more likely to be influenced by adjectives, verbs and nouns, or *topic modelling* where numbers do not offer insight into the main theme. Additionally, *text classification* such as classification of documents into fiction vs non-fiction, poetry, etc. On the other hand, numbers are crucial to *Financial News Analysis*, *date and time analysis*, or *health and medical texts*.

With our BoW model by which we compare a query string to the four documents. By removing the numbers from the documents, we don't lose valuable information about its content.

- **Stemming and Lemmatization:** Both Stemming and Lemmatization strip the word down to its root's form. The difference is that stemming works with a set of rules such as prefix, suffix, and etc. This presents an issue that the output of stemming may not have any meaning, e.g., the word "*flies*" will be stemmed to "*fli*", while it will be lemmatized to "*fly*". This issue is not raised from lemmatization; stripping each word down to its meaningful dictionary form. So, "*accepts*", "*accepted*", "*acceptance*", "*accepting*" will be reduced to "*accept*".

Benefits of stemming and lemmatization includes reduction of dimension, improved text analysis, enhanced search algorithms, upscaling information retrieval, or facilitating sentiment analysis. However, it can cause the "loss of meaning" issue where the 'stripped' words might not convey the exact context or semantic nuances in the original text.

We did not perform stemming. But we applied lemmatization to the corpus in this assignment and it is most likely affected the Crime and Punishment book. This reduces the number of vocabularies, a reduced the dimension.

There are additional operations to help 'clean' the text further in the preprocessing step. However, they are not appropriate to this assignment so we did not perform any of them. The following are some of such operations.

- **Removal of duplicates:** A method that can be used when frequency is not as important in the model. This could be more relevant in a model checking existence of a text. Removing duplicates does not give different weights to the words, so they remain the same level of importance, just more computationally efficient.
 - **Named Entity Recognition (NER):** A method used after tokenizing the documents to look up tokens and match them to a category in real life. Ex "Person", "Location", "Organisation", "Date" etc. This increases the understanding of the words in a document.
 - **Part-of-Speech (POS):** A method to pair/label the tokens in a document with its syntax in sentence structure. The labelled word is an abbreviation, like "I" is labelled "PRP" for possessive pronoun, "Books" labelled "NNS" for noun plural. This is also an additional step that can be used to increase the understanding of the words/tokens in a document.
 - **N-grams:** A method to predict the next word based on the previous words/tokens. N-grams can capture words that occur together (n-number of instances). The model needs to be trained on several sentences before it will make a calculated probability on which word comes next. The technique can be used in auto completions on sentences or to some degree in grammatical corrections.
- **Step 3: Vectorization:** A Vector Space Model (VSM) to get the computer to make sense of the words in the documents. The model CountVectorizer (sklearn) is selected as it is straight forward and outputs frequencies of words in the document. There are other model options such as the TF-IDF (Term Frequency-Invers Document Frequency), which is a way to give weights regarding a word's importance. The choice of VSM depends on that the model is going to be used for.
 - **Step 4: Post-Processing:** An optional step depending upon the needs, and the objectives of the analysis, during which one can further refine these representations for a better performance in the tasks that follow.

While BoW works off of raw vocabulary frequencies. TF-IDF gives weights to the term based on its importance. Tokens that are frequent in the specific documents but rather rare in documents across the board gets more weights. This is opposite to common vocabularies that appear across many documents where they receive lower weights. Additionally, higher weights also given to terms that are distinct to the document. TF-IDF increases discriminatory power of the BoW representations making them more effective for document retrieval, clustering or classification.

Bags of Words and Their Limitations: While it is undeniable that the Bags of Words (BoW) model is powerful in natural language processing tasks. However, it is as crucial to address its inherent shortcomings. BoW models views and treats texts as unordered collections of words, it neglects the sequence and semantics that often carry critical contextual meanings that can be overlooked. Additionally, synonyms are treated as different terms, and the model doesn't capture polysemy, where a single word can have multiple meanings based on context.

Furthermore, the BoW representation is usually sparse (filled with loads of zeros) since each text contains only a small portion of the total vocabulary. This can translate into inefficiencies and high memory usage.

Due to the aforementioned limitations, a fundamental concept such as BoW should remain its importance in natural language processing, but more sophisticated models such as Word Embedding, TF-IDF are able to capture more nuances in the text data.

III. RESULTS AND DISCUSSION

The table below show the reduction of number of tokens after each preprocessing steps (showing only steps that result in the change of number of tokens).

Preprocessing steps	Tromsø	Oslo	NLP
Tokenize	57	56	85
Remove stop words	41	35	58
Remove symbols	29	28	51
Number of tokens	29	28	51

- The word “tromsø” is retained throughout this token cleaning process using English stop words because “tromsø” is not an English word, therefore is not included in the list of English stop words. So, by processing the text, the word “tromsø” is not recognized and thus retained.
- The character “Ø” might look strange from an English perspective. But it is a character in Norwegian and Danish, and is a part of Unicode standard character. Therefore, it is not removed when the ‘word.isalpha()’ is performed.
- Having created the BoW representations which is a vector representing the frequencies of the word in the corpus (combined texts). The screenshot shows a summary of the said BoW having Ndim = 2, with the shape of 4 (documents) and 10,140 columns (number of words in the corpus). The number represent the frequencies of each vocabulary as they appear in each text. As mentioned earlier, it is sparse (filled with a lot of zeros) as each text contains only a portion of words in the corpus.

```
BoW_Representation Ndim: 2
BoW_Representation Shape: (4, 8317)
BoW_Representation:
[[ 0  0  0 ...  0  0  0]
 [ 0  0  0 ...  0  0  0]
 [ 0  0  0 ...  0  0  0]
 [ 4  3 10 ...  1  1 86]]
```

Frequency: The figure below shows the top 10-word frequency for the four texts with “city” and “oslo” being the most frequent word for Tromsø and Oslo text, respectively. All other word appears only once in these two texts.

For NLP text (a description of a Text Mining Course), the first three words (“*analysis*”, “*technique*”, “*application*”) capture the essence of the course perfectly.

While in the case of the Crime and Punishment book it makes perfect sense that the last name of the main character “Raskolnikov” appeared most frequent throughout the book.

```
#####
Top 10 Word Frequencies for Tromsø Text:
city: 3
arctic: 1
attraction: 1
base: 1
beautiful: 1
centre: 1
chasing: 1
colourful: 1
fascinating: 1
fjord: 1
#####
Top 10 Word Frequencies for Oslo Text:
oslo: 3
around: 1
banking: 1
big: 1
city: 1
climate: 1
considered: 1
end: 1
global: 1
hub: 1

#####
Top 10 Word Frequencies for NLP Text:
analysis: 4
technique: 3
application: 2
course: 2
information: 2
mining: 2
text: 2
access: 1
aim: 1
better: 1
#####
Top 10 Word Frequencies for Crime and Punishment:
raskolnikov: 718
one: 625
know: 573
would: 571
said: 518
could: 489
come: 466
like: 455
man: 454
time: 441
```

Similarity of a query string to the four texts: Two query strings to measure its similarity to the four texts. One is a description of the city of Bergen, the other is a few passages from a children book. A function was created to run the comparison using `cosine_similarity`. The result is the description of Bergen is most similar to the Tromsø text, while the passages from the book is most similar to the Crime and Punishment book.

There are other methods to compare the documents in terms of its similarity. The determinant factor for which one is more appropriate is the goals and the nature of such comparison. TF-IDF and Word Embedding are good alternatives to Cosine similarity.

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CONCLUSION AND FUTURE USE OF THE KNOWLEDGE

The knowledge gained from completing this assignment and its analysis helps to clarify a fundamental knowledge and solidify the understanding of not only text preprocessing, but also of text analysis, as well as fundamental methods and steps of NLP.

This knowledge we can use to build models that uses NLP. Either as a model that makes (human) unstructured text readable for a machine (NLU) or to generate a human-like response from a chatbot. It's a valuable knowledge to use while working with sentiment analysis, language processing when creating chatbots, or on research papers while working with text mining, topic modelling etc. Additionally, the knowledge and understanding can be used in creating spam filter or an IT-ticket system regarding priority of ticket.

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